

## 矩陣相乘(佔分 20 分)

### 問題描述：

將  $m \times n$  矩陣與  $n \times p$  矩陣相乘，會得到  $m \times p$  矩陣，如下圖所示。

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mp} \end{bmatrix}$$

矩陣內元素的運算法則如下：

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}, \text{ 其中 } 1 \leq i \leq m, 1 \leq j \leq p。$$

試撰寫程式，分別讀取  $a$  與  $b$  整數矩陣，矩陣相乘後輸出  $c$  整數矩陣的內容。

### 輸入說明：

第 1 行輸入整數陣列  $[a_{ij}]_{m \times n}$ ，其中  $1 \leq i \leq m, 1 \leq j \leq n, -999 \leq \{a_{ij}\} \leq 999, 1 \leq \{m, n\} \leq 20$ 。

第 2 行輸入整數陣列  $[b_{ij}]_{n \times p}$ ，其中  $1 \leq i \leq n, 1 \leq j \leq p, -999 \leq \{b_{ij}\} \leq 999, 1 \leq \{n, p\} \leq 20$ 。

### 輸出說明：

輸出整數矩陣  $a$  乘上整數矩陣  $b$  後的整數矩陣  $c$  的內容。

### 範例 1：

輸入：

$\{\{1, 2\}, \{3, 4\}\}$

$\{\{-1, 10\}, \{-9, 0\}\}$

輸出：

$\{\{-19, 10\}, \{-39, 30\}\}$

### 範例 2：

輸入：

$\{\{10\}, \{-2\}\}$

$\{5, 0, -5\}$

輸出：

$\{\{50, 0, -50\}, \{-10, 0, 10\}\}$

## Matrix Multiplication (20 points)

### Problem Description:

Multiplying an  $m \times n$  matrix by an  $n \times p$  results in an  $m \times p$  matrix.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mp} \end{bmatrix}$$

The calculation rule for the elements within the matrix is as follows:

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

Where  $1 \leq i \leq m$  and  $1 \leq j \leq p$ .

Write a program to read integer matrices  $a$  and  $b$ , and output the contents of the resulting integer matrix  $c$  after multiplication.

### Input Description:

Line 1: An integer matrix  $[a_{ij}]_{m \times n}$ , where  $1 \leq i \leq m$ ,  $1 \leq j \leq n$ ,  $-999 \leq \{a_{ij}\} \leq 999$  and  $1 \leq \{m, n\} \leq 20$ .

Line 2: An integer matrix  $[b_{ij}]_{n \times p}$ , where  $1 \leq i \leq n$ ,  $1 \leq j \leq p$ ,  $-999 \leq \{b_{ij}\} \leq 999$  and  $1 \leq \{n, p\} \leq 20$ .

### Output Description:

Output the contents of integer matrix  $c$  (the result of matrix  $a$  multiplied by matrix  $b$ ).

#### Example 1:

##### Input:

$\{\{1, 2\}, \{3, 4\}\}$

$\{\{-1, 10\}, \{-9, 0\}\}$

##### Output:

$\{\{-19, 10\}, \{-39, 30\}\}$

#### Example 2:

##### Input:

$\{\{10\}, \{-2\}\}$

$\{5, 0, -5\}$

##### Output:

$\{\{50, 0, -50\}, \{-10, 0, 10\}\}$